

Uddin, Md. Borhan

Ph.D. Candidate, School of Computer and Software, Shenzhen University

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RESEARCH PROFILE

- Ph.D. candidate in Computer Science and Technology at Shenzhen University, China, specializing in Artificial Intelligence, Deep Reinforcement Learning, and Simulation-to-Reality (Sim2Real) policy transfer for safety-critical embodied autonomy across UAVs, autonomous driving, robotic manipulation, and humanoid robotics.
- Experienced researcher, trainer, and educator with a strong academic foundation in Computer Science and Engineering and substantial professional experience in technical training, curriculum development, training needs analysis (TNA), performance evaluation, and team coordination.
- Proficient in C++, Python, Java, SQL, ROS/ROS 2, NVIDIA Isaac Sim/Isaac Lab, CARLA, Flightmare, Gazebo, and related AI/robotics simulation tools, with hands-on research experience in DRL-based policy learning, Sim2Sim transfer, runtime verification, and Sim2Real robotic systems.

OBJECTIVE

To pursue an academic career combining impactful research, high-quality teaching, and student mentorship, while contributing to innovation in intelligent autonomous systems and educating future computing professionals.

KEY PROFICIENCIES / RESEARCH EXPERTISE

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| ▪ Deep Reinforcement Learning & Policy Optimization | ▪ Python, C++, ROS 2 & AI/Robotics Toolchains |
| ▪ Sim2Real / Sim2Sim Policy Transfer | ▪ Research Design, Data Analysis & Reproducibility |
| ▪ Embodied AI & Autonomous Robotic Systems | ▪ Academic Writing, Publication & Peer Review |
| ▪ UAV, Autonomous Driving & Humanoid Robotics | ▪ Teaching, Mentoring & Technical Training |
| ▪ Robotics Simulation & Experimental Validation | ▪ Research Collaboration & Project Coordination |

ACADEMIC QUALIFICATIONS

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|----|---|--|
| 01 | Shenzhen University , Shenzhen, China
Ph.D. Candidate, School of Computer and Software | CGPA: 3.18/4.0
Sep 2022 – Present Expected Dec 2026 |
| | <ul style="list-style-type: none">• Advisor : Dr. Chen Jie• Dissertation Title : Reliable Sim2Real Policy Transfer for Safety-Critical Embodied Autonomy | |
| 02 | United International University , Dhaka, Bangladesh
M.Sc. in Computer Science and Engineering | CGPA: 4.0/4.0
May 2022 – Apr 2022 |
| | <ul style="list-style-type: none">• Advisor : Dr. A.K.M. Muzahidul Islam• Dissertation Title : Intelligent Patient Monitoring Using Digital Twins and Internet of Things | |
| 03 | Daffodil International University , Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering | CGPA: 3.54/4.0
May 2014 – Jun 2017 |
| | <ul style="list-style-type: none">• Advisor : Dr. Syed Akhter Hossain• Project Title : HeadTRAX: Web-Based Student-Teacher Communication and Appointment System | |
| 04 | Shahid Muktijodhya Mahabidyalaya , Jessore Board, Bangladesh
HSC in Science | GPA: 5.0/5.0
Jun 2009 – Jul 2011 |
| 05 | Samakal Secondary Bidyapith , Jessore Board, Bangladesh
SSC in Science | GPA: 4.81/5.0
Jan 2007 – May 2009 |

PUBLICATIONS

- | | |
|----|---|
| 01 | U. M. Borhan, A. Raza, T. Uddin, M. Islam, A. K. M. Muzahidul Islam, and J. Chen, "Performance, Adaptability, and Security in AI-Driven IIoT: A Survey," IEEE Internet of Things Journal, 2026, doi: 10.1109/JIOT.2026.3701588. |
| 02 | U. M. Borhan, A. Raza, Z. Lin, L. Wang, J. Li, and J. Chen, "Reliable Policy Transfer for Safety-Aware End-to-End Driving with Deep Reinforcement Learning," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026), Denver, Colorado, USA, June 2026, pp. 32134–32143. |
| 03 | A. Raza, U. M. Borhan, Y. Che, J. Chen, and L. Wang, "EdgeSAC: Graph Neural Soft Actor-Critic for Hierarchical IoV Resource Management," IEEE Transactions on Mobile Computing, 2026, doi: 10.1109/TMC.2026.3691088. |

- 04 B. Du, **U. M. Borhan**, T. Chen, J. Chen, J. Li, and J. Chen, "Robust UAV Policy Learning for Urban Infrastructure Surface Screening," in 2024 International Conference on Advanced Robotics and Mechatronics (ICARM 2024), Tokyo, Japan, 2024, pp. 1–8, doi: 10.1109/ICARM62033.2024.10715841.
- 05 J. Chen, W. Du, J. Lin, **U. M. Borhan**, Y. Lin, and B. Du, "Emergency UAV Landing on Unknown Field Using Depth-Enhanced Graph Structure," IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 4434–4445, 2025, doi: 10.1109/TASE.2024.3391017.
- 06 W. Du, J. Lin, B. Du, **U. M. Borhan**, J. Li, and J. Chen, "Autonomous Emergency Landing on 3D Terrains Approaches for Monocular Vision-Based UAVs," in 2023 International Conference on Advanced Robotics and Mechatronics (ICARM 2023), Sanya, China, 2023, pp. 105–112, doi: 10.1109/ICARM58088.2023.10218933.
- 07 T. Uddin, **U. M. Borhan**, A. K. M. Muzahidul Islam, S. Islam, and S. Shatabda, "Application of Internet of Things for Early Detection of COVID-19 Using Wearables," in 2021 International Conference on Software Engineering & Computer Systems and 4th International Conference on Computational Science and Information Management (ICSECS-ICOCSIM 2021), Pekan, Malaysia, 2021, pp. 405–410, doi: 10.1109/ICSECS52883.2021.00080.
- 08 **U. M. Borhan** and T. Uddin, "Solving Constraint Satisfaction Problem in TSP Using GA and DFS Algorithms," International Journal of Emerging Technologies and Innovative Research, vol. 7, no. 11, pp. 417–421, November 2020.
- 09 S. Ahmmed and **U. M. Borhan**, "A Study on Mamdani Fuzzy Logic to Implement the Programs of Washing Machine," International Journal of Emerging Technologies and Innovative Research, vol. 7, no. 10, pp. 3734–3738, October 2020.

MANUSCRIPTS UNDER REVIEW / IN PREPARATION FOR RESUBMISSION

- 01 **U. M. Borhan**, A. Raza, P. Haimei, and J. Chen, "Omni4D: Object-Centric Policy Transfer for Humanoid Manipulation," submitted to The Forty-First AAAI Conference on Artificial Intelligence (AAAI 2027).
- 02 **U. M. Borhan**, A. Raza, Z. Lin, J. Peng, J. Li, and J. Chen, "Ego-Relational Policy Transfer for Safety-Aware End-to-End Autonomous Driving," submitted to IEEE Transactions on Pattern Analysis and Machine Intelligence.
- 03 **U. M. Borhan**, B. Du, A. Raza, J. Li, and J. Chen, "Autonomous Policy Transfer for GPS-Denied UAV Infrastructure Inspection," submitted to IEEE Transactions on Intelligent Vehicles.
- 04 **U. M. Borhan**, A. Raza, B. Lv, J. Li, and J. Chen, "Robust Speed Control for UAV Infrastructure Inspection Under Visual Localization Degradation," submitted to IEEE Transactions on Automation Science and Engineering.
- 05 A. Raza, **U. M. Borhan**, M. W. A. Ashraf, M. M. Abro, J. Chen, and L. Wang, "Edge Intelligence for Resource Management in Hybrid RF-VLC Internet of Vehicles," submitted to IEEE Internet of Things Journal.
- 06 A. Raza, **U. M. Borhan**, A. Nasir, J. Li, J. Chen, and L. Wang, "Joint Radio-Compute Resource Management for Clustered Vehicular Edge Networks," submitted to IEEE Transactions on Mobile Computing.
- 07 Sameera Sameera, **U. M. Borhan**, A. Raza, Q. Liu, and K. Sharif, "Adaptive Post-Quantum Authentication for SDN-Managed Smart-Home IoT," submitted to IEEE Transactions on Dependable and Secure Computing.
- 08 **U. M. Borhan**, Sameera Sameera, A. Raza, F. Saeed, and K. Sharif, "Federated Risk Learning for Adaptive Post-Quantum IoT Authentication," submitted to The Fifteenth International Conference on Learning Representations (ICLR 2027).
- 09 A. Raza, **U. M. Borhan**, Sameera Sameera, L. Ouyang, and J. Chen, "Selective Feature-Node Transformation for Heterophilic Graph Learning," submitted to The Fifteenth International Conference on Learning Representations (ICLR 2027).
- 10 Sameera Sameera, A. Raza, R. Ali, and **U. M. Borhan**, "BCPT-Med: Credal Bounded-Consensus Partial Transport for Imbalanced Medical Image Clustering," submitted to The Forty-First AAAI Conference on Artificial Intelligence (AAAI 2027).
- 11 A. Raza, **U. M. Borhan**, Y. Yu, and J. Chen, "Conformal Safety Filtering under Enforcement Mismatch," submitted to The Forty-First AAAI Conference on Artificial Intelligence (AAAI 2027).
- 12 Z. Lin, Z. Chen, **U. M. Borhan**, and J. Li, "Distilling Reliable Guidance from Suboptimal Trajectories via Hierarchical Expert-Calibrated Alignment," submitted to The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026).
- 13 L. Al Amin, **U. M. Borhan**, M. Islam, M. M. J. Ayan, G. M. I. Mahmud, M. B. Shahid, S. Bhowmick, R. K. Das, A. A. Bondhon, T. Uddin, A. K. M. Muzahidul Islam, and T. T. Nguyen, "Artificial Intelligence and Machine Learning in Combating COVID-19: Lessons Learned for Future Pandemics for South Asia," submitted to Artificial Intelligence in Health, Review Article.

ACADEMIC AND PROFESSIONAL EXPERIENCE

- 01 **Shenzhen University**, Shenzhen, China
Graduate Teaching Assistant, School of Artificial Intelligence
Responsibilities and Achievement:



Mar 2024 - Present

- Assisted with teaching, student support, and academic activities, including coursework guidance, programming, technical problem solving, and research-oriented learning.
- Contributed to academic mentoring, research communication, and student engagement alongside doctoral research in AI, deep reinforcement learning, and autonomous systems.

02 New Horizons, Dhaka, Bangladesh

Training Manager and Senior Instructor



Responsibilities and Achievements:

Oct 2018 - Dec 2022

- Managed technical-training programs, including curriculum development, instructional planning, trainer coordination, training needs analysis, performance evaluation, and quality assurance.
- Delivered and coordinated programming and application-development training for 300+ batches and 4,500+ learners, while supporting training materials and learner progress evaluation.

03 Skill Jobs, Dhaka, Bangladesh

Assistant Coordinator, Training and Development



Responsibilities and Achievements:

Oct 2017 - Sep 2018

- Coordinated organization-wide training programs, including training needs analysis, instructional planning, participant coordination, and selection of appropriate training methods and activities.
- Contributed to a 20% revenue increase within six months by supporting the planning, development, and implementation of annual professional training programs.

04 New Horizons, Dhaka, Bangladesh

Applications Instructor, Training Department



Responsibilities and Achievements:

Nov 2015 - Sep 2017

- Delivered professional training in programming, application development, databases, and productivity technologies for diverse learner groups and professional participants.
- Monitored learner progress, performance, and training metrics, while updating course content to remain aligned with evolving industry requirements and technologies.

AWARDS, HONORS, AND PARTICIPATIONS

- Attendance Certificate, IEEE/CVF Conference on CVPR 2026), Jun 5-7, 2026
- Volunteer Award, Shenzhen Government, China, Jun 2025.
- Best International Volunteer, Nanshan District International Volunteer Services, Shenzhen, China, Dec 2024.
- Shenzhen Universiade International Scholarship, Shenzhen, China, 2024.
- Certificate of Participation, 9th IEEE ICARM, Tokyo, Japan, Jul 8-10, 2024.
- Best Conference Paper Finalist, 8th IEEE ICARM, Sanya, China, Jul 8-10, 2023.
- Shenzhen Universiade International Scholarship, Shenzhen, China, 2023.
- Best Trainer Award, New Horizons Worldwide International Conference, Dec 2022.
- Best Paper Award, ICSECS-ICOCSIM, Malaysia, Aug 2021.
- Best Innovative Trainer, New Horizons Bangladesh, Jan 2021.
- Youth Leadership Award, GREEN CROSS and COP21, Oct 2020.
- Book Reading Competition Award, Bishwo Shahitto Kendro, Mar 2011.

ACADEMIC EXCHANGE PROGRAMS

01	University of Toronto, Toronto, Canada	Final Letter Grade: A+
	English for Academic Communication, School of Continuing Studies	Mar 2023 – Jun 2023
02	National Youth Services Council, Maharagama, Sri Lanka	N/A
	5th Bangladesh and Sri Lanka Youth Exchange Program	Jan 2017 – Feb 2017
03	Sias University, Henan, China	CGPA: 3.94/4.0
	Chinese Language and Culture, Sias Winter Camp	Oct 2016 – Mar 2017
04	University Malaysia Perlis, Perlis, Malaysia	CGPA: 3.25/4.0
	Engineering Entrepreneurship; Global Sustainability & IT, Asian Summer Program	Aug 2014 – Nov 2014

TECHNICAL TRAINING EXPERTISE

▪ Programming: C++, Java SE & EE, Python ▪ Artificial Intelligence: ML, Neural Networks, NLP ▪ Data Structures: Arrays, Trees, Graphs, Sorting ▪ Project Management: Microsoft Project	▪ Web Development: HTML, CSS, JS, PHP, Laravel ▪ Frameworks & APIs: Spring Boot, Django, REST APIs ▪ Database Management: MySQL, Microsoft Access ▪ Productivity & Tools: Word, Excel, PowerPoint, Unity
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CORPORATE TRAINING EXPERIENCE

- Advanced Microsoft Excel, Texeurop (BD) Ltd., Gazipur, Bangladesh, 2022.
- Artificial Intelligence, Bangladesh Army Information Technology Support Organizations (AITSO), 2021.

- Productivity Tools with Microsoft Excel, Transcom Beverages Ltd., 2021–2022.
- Advanced Microsoft PowerPoint, Transcom Beverages Ltd., 2021.
- Microsoft Project, Meghna Group of Industries, Dhaka, Bangladesh, 2021.
- Microsoft Project Professional, China Civil Engineering Construction Corporation (CCECC), Payra Port, 2020.
- Certified Internet Web Professional (CIW), Hamdard University, Bangladesh, 2018–2019.
- Programming in C++, Military Institute of Science and Technology, Dhaka, Bangladesh, 2017.
- MySQL Database Administration, Ministry of Labor and Employment, Bangladesh, 2017.

PROFESSIONAL MEMBERSHIPS AND SERVICE

- Graduate Student Member (97043699), R10 -Asia and Pacific Region, Guangzhou Section, IEEE.
- Member (M-49934), Institution of Engineers, Bangladesh (IEB).
- Director, YES Entrepreneur Bangladesh Limited.
- Former Rover Mate, Air Region, Bangladesh Scouts.
- Volunteer, Youth Against Hunger Bangladesh, Mastul, and Asian Network of Youth Volunteers Society.

LEADERSHIP AND ADDITIONAL ACTIVITIES

- Team Leader, SZU-PIONEER, 2026 Global Humanoid Robot Challenge (GHRC); Apr – Aug 2026.
- Scout Unit Leader Basic Course, Bangladesh Scouts, Mar 2019.
- Career in Global Marketplace, organized by Jobsbd and the British Council, Mar 2018.
- CISCO and ORACLE Training Program, organized by Jobsbd and Daffodil Int'l University, Feb 2018.
- Youth Parliamentarian in the 3rd Youth Parliament on Right to Work, Mar 2016.
- Youth Parliamentarian in the 2nd Youth Parliament on Right to Food, Oct 2015.
- Bangla Translate-a-thon, organized by GDG Bangla, May 2015.

CERTIFICATIONS

- | | |
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| ▪ Advanced Programming in C++ | ▪ Cybersecurity Essentials |
| ▪ Programming Essentials in Python | ▪ Mobile Sites Certification and Digital Sales |
| ▪ Introduction to Cybersecurity | ▪ AdWords and Analytics Certifications |

PROFESSIONAL DEVELOPMENT AND WORKSHOPS

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| • Java SE 7 Programming I & II | • Digital Marketing for IT Professionals |
| • Web Application Development | • Multimedia and Graphic Design |
| • Mobile Application and Game Development | • News Presentation and Reporting |

PERSONAL DETAILS

Name (Surname, Given Name)	: UDDIN, MD BORHAN	National ID	: 2814010589
Father's Name	: Md. Abu Bakkar Biswas	Passport Number	: A13868419
Mother's Name	: Nazira Begum	Date of Birth	: Jan 02, 1994
Spouse's Name	: Tahmina Karim	Nationality	: Bangladesh
Permanent Address	: 773, Sikandar Abu Zafar Road, Jatpur, Tala, Satkhira, Bangladesh-9420		

LANGUAGES

Bangla: Native | English: Fluent | Chinese: Beginner | Urdu: Beginner | Hindi: Beginner

REFERENCES

Dr. Chen Jie Distinguished Research Fellow & Associate Professor School of Artificial Intelligence, Shenzhen University, China Contact: +86 17841-021782, chenjie@szu.edu.cn	Dr. A.K.M. Muzahidul Islam Professor, School of Computing and Informatics Albukhary International University, Malaysia Contact: +880 1754-165915, muzahidul.islam@aiu.edu.my
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I hereby declare that the information provided above is true and accurate to the best of my knowledge and belief. I accept responsibility for any errors or inaccuracies that may be identified.



Uddin, Md. Borhan
(Sep 25, 2026)